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Review article

Committee of the new positions of the Société française de radiothérapie oncologique: Toward an advanced practice master in radiotherapy



Commission des nouveaux statuts de la Société française de radiothérapie oncologique : vers un master de pratique avancée en radiothérapie

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ABSTRACT

In 2022, the radiation therapist committee of the Association française du personnel paramédical d'électroradiologie médicale (AFPPE, French association of paramedical electroradiology technicians), the Société française de radiothérapie oncologique (SFRO, French society of radiation oncology) and the Syndicat national des radiothérapeutes oncologues (SNRO, national syndicate of radiation oncologists) have been committed to working on the development of advanced practice roles. The objective of this article is to report the activities that should be in the scope of radiation therapists advanced practice and describe the competences required for these activities. This work was carried out by six radiation therapists, six radiation oncologists and one medical physicist representatives of the French national societies for each professional group. First, a basic list of activities was established and then competences were identified for groups of activities. In total, the list includes five core competences, nine competences and nine groups of activities that can be divided into the four pillars of advanced practice. The nine groups of activities can be presented in seven different dimensions including patient care and support, treatment planning, treatment imaging and delivery, management and consultancy, quality and risk management, research and innovation, education and training. The French advanced practice competences framework was developed with a multidisciplinary group to move forward the project of a master degree in advanced practice in radiation therapy in France.

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R É S U M É

En 2022, la commission radiothérapie de l'Association française du personnel paramédical d'électroradiologie médicale (AFPPE), la Société française de radiothérapie oncologique (SFRO) et le Syndicat national des radiothérapeutes oncologues (SNRO) se sont engagés dans le développement de la pratique avancée. L'objectif de cet article est de rapporter les activités qui devraient faire partie du champ de pratique d'un manipulateur d'électroradiologie médicale de pratique avancée en radiothérapie et de décrire les compétences requises pour ces activités. Ce travail a été réalisé par six manipulateurs

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d'électroradiologie médicale, six oncologues-radiothérapeutes et un physicien médical, tous membres des sociétés savantes nationales ou associations de chaque groupe de professionnels. Tout d'abord, une liste basique des activités a été établie, puis des compétences ont été identifiées pour les groupes d'activités. En tout, la liste comprend cinq blocs de compétences, neuf compétences et neuf groupes d'activités qui peuvent être répartis dans les quatre piliers de la pratique avancée. Les neuf groupes d'activités peuvent être présentés dans sept dimensions différentes incluant les soins au patient et son soutien, la planification de traitement, l'imagerie de traitement et la délivrance, le suivi du patient et les conseils, la qualité et la gestion des risques, la recherche et l'innovation, l'enseignement et la formation.

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1. Introduction

Global cancer statistics show an increase of cancer incidence [1]. Moreover, technological developments such as stereotactic radiation therapy and hypofractionated radiation therapy have the effect of treating more patients including those that were previously deemed untreatable [2]. These changes result in a greater number of steps in radiation therapy preparation (i.e. initial consultation, CT scan, organs at risk and target volume delineation) as well as in patient follow up [3]. In addition, the current regulations defining both the role of each staff member and a specific educational framework, can limit access to innovation. For example, in France, validation of positioning imaging for radiation therapy treatment delivering more than 8 Gy must be done by a radiation oncologist. This also applies to re-delineation of organs at risk and target volume in a context of stereotactic treatment and adaptive radiotherapy respectively, despite the implementation of artificial intelligence tools and specific education for radiation therapists. In addition, the increase in the number of patients' demands lead to a higher volume of patients followed up weekly by radiation oncologists. Increased workload on the radiation oncologists mainly might be a significant cause of stress, with a higher risk of errors and a reduction of the performance [4].

To face this situation, one solution could be for radiation therapists to work across boundaries with a specific educational program on new tasks that are currently considered out of their scope of practice. Task shifting is a method that can be used to delegate specific tasks from physician(s) to non-medical healthcare professionals with the aim to optimise the patient's care pathway and patient workflow. In France, task shifting must be framed by a cooperation protocol registered with the Agence régionale de santé (ARS) [5,6].

More than a simple task shifting, advanced practice describes a professional working outside of their scope of practice and demonstrating expertise in a specific area. They work autonomously, taking on a leadership role in the development of radiotherapy services, and research associated with their own specialty [7]. European qualifications framework (EQF) level 7 or 8 is recommended for advanced practitioners, corresponding to at least master degree level [7]. Advanced practice roles have been successfully already and implemented widely in other healthcare professions and have shown benefits in terms of patient care and access to healthcare services [8,9]. In France, advanced practice is recognised for nurses with a master degree covering various field of expertise including oncology and haemato-oncology [10]. However, it has not been yet possible for French radiation therapists to obtain a master degree for advanced practice in radiation therapy and practice as an advanced practitioner.

The Association française du personnel paramédical d'électroradiologie médicale (AFPPE) is the French association of paramedical electroradiology technicians representative of

radiographers for all fields of practice including radiation therapy, diagnostic imaging, nuclear medicine, interventional radiology and electrophysiology. All French radiographers are graduated with a combined degree recognised EQF 6. The term radiation therapist was specifically used in this article to speak about radiographers working in radiation therapy.

The radiation therapist committee of the AFPPE, the Société française de radiothérapie oncologique (SFRO, French society of radiation oncology) and the Syndicat national des radiothérapeutes oncologue (SNRO, French national syndicate of radiation oncologists) have been committed to working on the development of advanced practice roles in radiation therapy. In 2022, representatives of these societies started working to define the groups of activities that should be in the scope of practice of advanced practice RTTs, and to describe the competences required for these groups of activities. The objective of this article is to report this work.

2. Material and method

This work was carried out jointly by SFRO, SNRO and AFPPE as part of the Committee of the new positions of SFRO. This committee includes 12 members, six radiation oncologists and six radiation therapists. Additionally, the Société française de physique médicale (SFPMP, French society of medical physics) was solicited to contribute to this work.

First, a basic list of the activities was done by every member based on and discussed during meetings. In line with public health needs and the level of education of radiation therapists in France, as well as the European Society for Radiotherapy and Oncology (ESTRO) position on advanced practice, a triage based on topics was done [7]. Secondly, based on the French competences referential for the advanced practice nurses, competences were identified for each activity resulting from the first stage [8]. The competences identified in this work overlap with the competency framework for French advanced practice nurses in oncology and haemato-oncology.

Finally, this list of activities and competences for the advanced practice radiation therapists was shared and discussed between the Committee of the new positions and the Comité de réflexion pour l'évolution du métier de manipulateur d'électroradiologie médicale (CREM, professional progression review committee) in 2023, meetings in which members of the radiation therapy committee of the AFPPE participated. Launched on the initiative of the Comité national professionnel des manipulateurs d'électroradiologie médicale (CNPMM, national professional council of medical electroradiology technicians) with the aim of positioning several new health professions as a progression of the radiographer/radiation therapist profession, the CREM brought together expert radiographers in various field of practice: education, radiation therapy, diagnostic imaging, nuclear medicine and

Table 1

French advanced practice competences framework elaborated by the multidisciplinary team of the new position committee of the Société française de radiothérapie oncologique (SFRO, French society of radiation oncology).

Core competence	Competence	Groups of activities
Implementation of complex diagnostic, therapeutic procedures and techniques that contribute to the care pathway	Assess the state of health of patients as part of recourse motives to medical consultations for cancer pathologies Define and implement patients' care journeys based on the global assessment of their state of health	1. Data observation, collection and interpretation in the course of diagnostic or therapeutic interventions 2. Prescription, prescription renewal and performance of technical acts adapted to the recourse motive, the pathology and the person's situation 3. Carrying out medicotechnical and care activities for diagnostic and/or therapeutic aims
Organization and coordination of patient care and education activities	Implement techniques, care and medico-technical practices for diagnostic and/or therapeutic purposes Organise and coordinate care and medico-technical activities within the framework of a care and health pathway in collaboration with all the actors concerned Design and implement prevention and therapeutic education actions	4. Organization and coordination of care activities and medicotechnical interventions, the patient's care and health pathway 5. Design, implementation and evaluation of prevention and therapeutic education actions
Implementation of the quality approach and risk prevention	Conduct quality, risk management and prevention monitoring in the different sectors	6. Development and implementation of quality, risk management and risk prevention monitoring related to care, heavy equipment, and medical devices
Information and training of professionals in the related sectors	Implement training and information actions	7. Implementation of training and information actions for multidisciplinary teams
Professional intelligence, research work and practice improvement initiatives	Implement and conduct actions to evaluate professional practices in view of technical and regulatory developments Research, analyse and produce professional and scientific data.	8. Implementation and evaluation of actions to improve professional practices 9. Contribution and conduct of studies and research work

interventional radiology. The competences were also classified in the four pillars of the advanced practice described in the literature and posteriori, in the seven dimensions used by Oliveira et al. [8,9,11].

3. Results

French advanced practice competences framework elaborated by the multidisciplinary team of the new position committee of the SFRO is presented in Table 1. The work undertaken allowed to identify five core competences from which nine groups of activities were defined related to the implementation of procedures in the field of radiation therapy. These competences and groups of activities are necessary for the treatment and monitoring of pathologies taken care by radiation therapy oncology and for the organisation and coordination of patient care pathways, while ensuring the improvement of practices.

Implementing complex acts and techniques for diagnosis and therapy in the patient care pathway could strengthen clinical patient care by ensuring that these actions are complementary and gradable. Radiation therapist competences should be strengthened in the context of complex treatments requiring complementary competences, such as validation of gross tumour volume delineation or treatment planning validation for adaptive radiotherapy.

Defining health care pathways, as care is being adapted to hazards management and treatment interruptions, could be strengthened by the organisation and coordination of patient care and education activities. Attention is paid in particular to manage computerised pathways, monitor and adapt workflows meeting suited to the patient's personalised needs, monitor coordination of follow-up postponements elements and cancel file preparation. In the same way, the quality and risk prevention approach that is the basis for the safety elements and definition of operating processes in radiation therapy, could facilitate the implementation of clinical paramedical audits and a proactive approach to risk prevention. This core competence should allow for structuring learning on data management, in particular regarding the collection and

data integration such as patient-reported experience and outcome measures.

Informing and training professionals in the related sectors is crucial to facilitate their intervention in the facilities providing initial education and to foster continuing professional development both for their peers and for multidisciplinary teams. It also helps supporting their professional leadership as an actor in the prevention of cancer and as a professional who play a crucial role to educate patients. Research work and the evaluation of practices are integrated to promote innovation and excellence in care. The competences can also be presented for each of the four pillars of advanced practice as illustrated in Table 2 and the activities can be divided into seven dimensions as shown in Table 3.

4. Discussion

As advanced practice roles required at least an EQF 7 corresponding to a master degree level, the multidisciplinary team of the SFRO new position committee worked on the groups of activities of an advanced practitioner and the competences required to perform these groups of activities in the context of radiation therapy. This document, presented in the form of a French advanced practice competences framework, constitutes a first step for the education program of an advanced practice master in radiation therapy in France. The core competences and competences of the framework are based on four pillars of the advanced practice described in the literature including:

- clinical and expert practice;
- professional leadership and consultancy;
- education, training and development;
- practice and service development, research and evaluation [7,9,11].

Recent reviews described and evaluated advanced practice roles, including various activities in different clinical areas or site specific [8,9]. In 2021, 87 published studies described advanced practice roles including activities and their impact. As Oliveira et al.

Table 2
Competences of the French advanced practice competences framework elaborated by the multidisciplinary team of the new position committee of the Société française de radiothérapie oncologique (SFRO, French society of radiation oncology) presented for each of the four pillars of advanced practice.

Pillars of advanced practice	Competences
Clinical and expert practice	To assess the state of patient health as part of recourse motives to medical consultations for cancer pathologies Define and implement patients' care plan based on the global assessment of their state of health Implement techniques, care and medicotechnical practices for diagnostic and/or therapeutic aims Design and implement prevention and therapeutic education actions
Professional leadership and consultancy	Organise and coordinate care and medicotechnical activities within the framework of a care and health pathway in collaboration with all the actors concerned Conduct quality, risk management and prevention monitoring in the different sectors Implement training and information actions
Education, training and development	
Practice and service development, research and evaluation	Implement and conduct actions to evaluate professional practices in view of technical and regulatory developments Research, analyse and produce professional and scientific data

Table 3
Development of advanced practice roles of radiation therapists in France: groups of activities presented in seven dimensions defined by the Association française du personnel paramédical d'électroradiologie médicale (AFPPE, French association of paramedical electroradiology technicians), the Société française de radiothérapie oncologique (SFRO, French society of radiation oncology) and the Syndicat national des radiothérapeutes oncologue (SNRO, French national syndicate of radiation oncologists).

Activities and tasks dimensions	Groups of activities of the French advanced practice competences framework
Patient care and support	Data observation, collection and interpretation in the context of diagnostic or therapeutic interventions Prescription, prescription renewal and performance of technical acts adapted to the reason for the appeal, the pathology and the situation of the person Design, implementation and evaluation of prevention and therapeutic education actions
Treatment planning	Carrying out medicotechnical activities and care for diagnostic and/or therapeutic aims
Treatment imaging and delivery	Carrying out medicotechnical and care activities for diagnostic and/or therapeutic aims
Management and consultancy	Organisation and coordination of care activities and medicotechnical interventions, the patient's care and health pathway
Quality and risk management	Development and implementation of quality, risk management and risk prevention approaches related to care, heavy equipment, and medical devices
Research and innovation	Implementation of evaluation actions to improve professional practices Contribution and conduct of studies and research works
Education and training	Implementation of training and information actions for multidisciplinary teams

mentioned, only 11 % of these roles relied fully on the four pillars of the advanced practice. A majority of these studies reported exclusively the clinical practice pillar. However, the education program's goals would enable the above-mentioned four pillars to be fully implemented, with a view to promoting competences at national

level, and harmonising them internationally [12]. Moreover, relying only on clinical practice would lead towards a specialisation of radiation therapists, like specialised nurses (i.e., in paediatrics or anaesthesia), and not towards a full advanced practice.

Our results were incorporated into the CREM's deliberations and discussed on 10 January 2024 at a meeting between the CNPMEM and the French ministry of health's Direction générale de l'offre de soins (DGOS). In line with the CREM's conclusions, the CNPMEM is in favour of developing an EQF 7 training programme dedicated to advanced practitioners [13]. Facing difficulties to offer a common master's-level education programme to all the radiographers/radiation therapists specialities in France, which are added both the need for radiographers to specialise and the advanced practice for radiation therapists, this work suggests to make a connection between the advanced practice nurses education programme and the project for an education programme for advanced practice radiation therapists. Consequently, the French advanced practice competences framework attempts to consider all four pillars.

The activities and tasks highlighted in this work can be grouped into seven dimensions: patient care and support; treatment planning; treatment imaging and delivery; management and consultancy; quality and risk management; research and innovation; education and training. These dimensions were used by Oliveira et al. in their systematic review for the triage of the studies on advanced practice roles and showed that advanced practitioners can play a role all along the radiation therapy care pathway as Harnett et al. suggested with the model of advanced practice radiation therapist developed in Ontario [8,14].

Furthermore, it can be seen that except on treatment planning and treatment imaging and delivering, a majority of these dimensions can be found in the referential of activities and competences for French advanced practice nurses in oncology and haematology [10]. It also important to emphasise that a demography of about 3000 radiation therapists in France for example, with professional shortage could be a barrier to implementing both advanced practitioners as well as an education program for future advanced practitioners in particular if we consider that all radiation therapists will not become advanced practitioners. Development of advanced practice is relevant for patient care, organisation of the radiation therapy departments, for attracting and retaining students and radiation therapists. Implementation of advanced practitioners is desired by French radiation therapists and radiation oncologists for all the reasons mentioned previously, but the radiation therapist shortage slowed this project as shown in two surveys sent to the French radiotherapy departments in 2022 [15,16]. Like advanced practice nurses, advanced practice radiation therapists will not replace the radiation oncologists or the residents in radiotherapy but these professionals should work together and learn from each other to fill the gaps found along a standard radiotherapy patient pathway [14]. This career progression is also an opportunity for these experienced radiation therapists to become trainers or for few of them a program leader in the context of initial education as recommended by International Atomic Energy Agency [17].

However, an educational program is not sufficient to implement advanced practice roles. After obtaining their master's degree, these professionals need to be entitled to practise as advanced practitioners. In France, health professions are regulated by a decree that sets out both a framework for their activities and the required education for this profession with the underlying competences that the professional will have acquired after graduation. Health professionals must not practise without this decree even if they obtained their degree.

With the goal to maintain and foster this momentum, a research project has been developed. This project aims to evaluate benefits of

task shifting that will be established by cooperation protocols registered with the ARS for France such as the local initiatives reported in 2023 [18]. The benefits will be evaluated in terms of patient care quality, patient satisfaction, waiting times, patient throughput, time savings and cost effectiveness from among the suggestions given by Harnett et al. [19]. The aim of this project is to highlight the dynamic of the radiation therapy professional community, to show that task shifting works and can be beneficial for patients and institutions and to emphasise that task shifting solely relying on cooperation protocols is insufficient, and thus, an advanced practice role is the solution.

5. Conclusion

In this work, the Committee of the new positions of the SFRO, SNRO and AFPPE presents the French advanced practice competences framework including five core competences, nine competences and nine groups of activities for advanced practice radiation therapists. This advanced practice competences framework was developed collegially and emphasises the importance of a holistic and multidisciplinary approach in patient care in radiation therapy and reflects a commitment to continuous improvement in the quality of care and patient safety. This work opens several perspectives for the education of radiation therapist advanced practice.

Author contributions

sb: conceptualization, methodology, resources, writing – original draft, visualization, supervision, project administration; SC, AB, ADO: conceptualization, methodology, writing – original draft; CB: review and editing.

Disclosure of interests

The authors did not disclose their relationships/activities/interests.

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